

Performance Analysis of 256-QAM and M-PSK Under AWGN and Phase Noise

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Abstract—This study presents a Simulink-based simulation of a digital communication system employing 256-QAM modulation under the effects of additive white Gaussian noise (AWGN) and phase noise. The simulation framework includes a complete modulation/demodulation chain, error rate computation, and constellation diagram analysis, enabling a detailed investigation of system performance. Results demonstrate that while 256-QAM offers high spectral efficiency, its performance is notably degraded by phase noise, as evidenced by increased bit error rates (BER) and dispersion in the constellation diagram. A comparison with M-PSK modulation further illustrates the trade-offs between spectral efficiency and noise resilience.

Index Terms—256-QAM, M-PSK, AWGN, phase noise, bit error rate,

I. INTRODUCTION

Digital communication systems play a vital role in modern telecommunications by transmitting information efficiently over various channels. Unlike analog communication, digital communication represents data using discrete signal levels, making it more robust to noise, interference, and signal degradation. One of the key modulation techniques in digital communication is Quadrature Amplitude Modulation (QAM), which combines both amplitude and phase modulation to improve spectral efficiency.

II. BACKGROUND INFORMATION

A. Quadrature Amplitude Modulation (QAM)

Quadrature Amplitude Modulation (QAM) is a widely used modulation scheme in modern wireless and wired communication systems, including Wi-Fi, 4G/5G networks, and cable television. QAM works by varying both the amplitude and phase of a carrier signal to encode digital data. In general, higher-order QAM schemes, such as 256-QAM, allow for higher data transmission rates by increasing the number of distinct symbols in the signal constellation.

B. 256-QAM Modulation

256-QAM is an advanced digital modulation scheme where each transmitted symbol represents 8 bits of data (since $2^8 = 256$). This high-order modulation technique provides improved data rates but requires a higher signal-to-noise ratio (SNR) for reliable detection due to reduced symbol spacing in the

constellation diagram. The 256-QAM constellation consists of 256 unique points, with each point corresponding to a specific combination of amplitude and phase values. The closer spacing of these points makes 256-QAM more susceptible to noise, distortion, and phase errors.

III. EFFECTS OF AWGN IN COMMUNICATION SYSTEMS

In practical communication channels, transmitted signals are affected by noise, which degrades signal quality. A common noise model used in digital communication system analysis is Additive White Gaussian Noise (AWGN). AWGN represents random noise with a constant power spectral density over all frequencies and follows a Gaussian distribution. The presence of AWGN in a communication system results in symbol errors, which can be quantified using the bit error rate (BER).

IV. PHASE NOISE AND ITS IMPACT ON QAM

In addition to AWGN, another key impairment in QAM-based systems is phase noise, which arises due to imperfections in oscillators and local carriers. Phase noise causes distortions in the received signal by introducing random variations in phase, leading to symbol misinterpretation in high-order QAM constellations like 256-QAM. Even small amounts of phase noise can significantly impact system performance, increasing BER and limiting system reliability.

V. SIMULATION USING SIMULINK

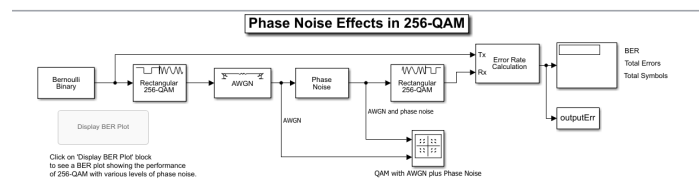


Fig. 1. Block diagram on simulink

- The Bernoulli Binary Generator block generates a random signal consisting of a sequence of 8-bit binary values in the range [0, 255].

- The Rectangular QAM Modulator Baseband block modulates the signal using baseband 256-ary QAM.
- The AWGN Channel block models a noisy channel by adding white Gaussian noise to the modulated signal.
- The Phase Noise block introduces noise in the angle of its complex input signal.
- The Rectangular QAM Demodulator Baseband block demodulates the signal.

These additional model blocks can help you interpret the simulation

- The Constellation Diagram block displays constellation diagram of the signal with AWGN and phase noise added.
- The Error Rate Calculation block counts bits that differ between the received signal and transmitted signal.
- The To Workspace block, labeled outputErr, outputs the results to the workspace for use when plotting the results.
- A Callback Button labeled Display BER Plot opens a plot showing the E_b/N_0 performance curves for 256-QAM transmission and reception at various levels of phase noise.

VI. SIMULINK SIMULATION RESULTS

A. Digital Modulation

The model simulates QAM, which is a method for converting a digital signal to a complex signal. The model modulates the signal onto a sequence of complex numbers that lie on a lattice of points in the complex plane, known as the constellation of the signal. A plot of these points is called a scatterplot or constellation diagram of the signal.

The constellation diagram shown here displays the baseband 256-ary QAM with AWGN added, and with AWGN and phase noise added. The points in the constellation diagram do not lie exactly on the constellation shown in the figure because of the added noise. Phase noise alters the angle of the complex modulated signal, causing a radial displacement of constellation points.

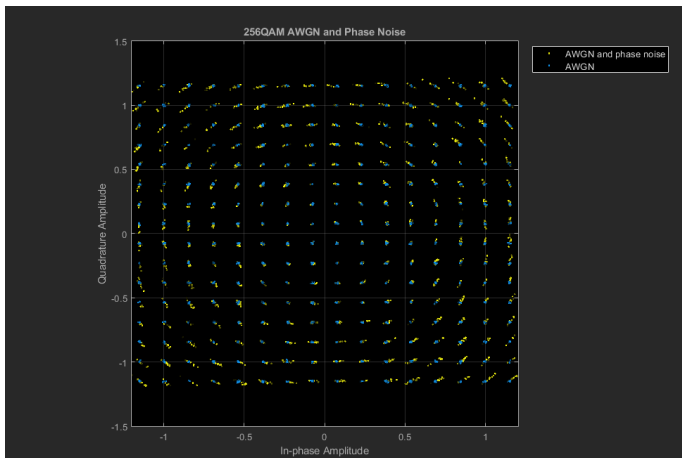


Fig. 2. 256QAM AWGN and Phase Noise

B. Display the Error Rate

The display error rate, typically measured as Bit Error Rate (BER) in a 256-QAM system, quantifies transmission errors due to noise and distortions. Factors like AWGN, phase noise, and closely spaced constellation points make 256-QAM susceptible to errors, especially in low SNR conditions. In Simulink, BER is computed by comparing transmitted and received bits, while the constellation diagram visually shows symbol deviations. Understanding BER helps optimize communication system performance by implementing noise mitigation and error correction techniques

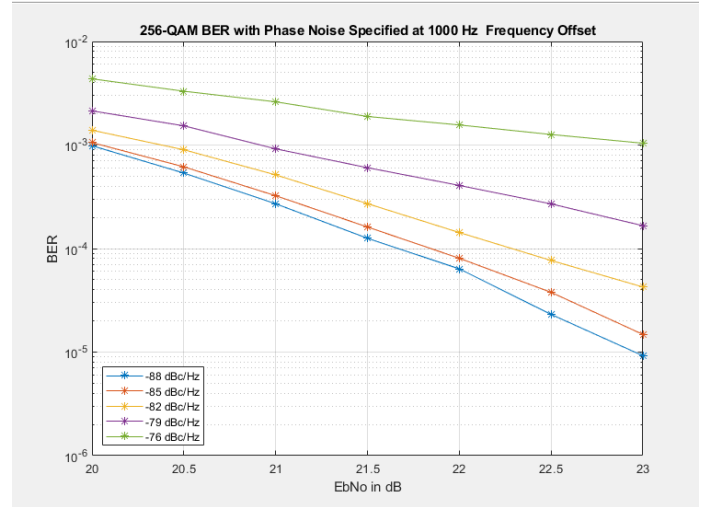


Fig. 3. 256QAM BER with Phase Noise Specified at 1000Hz Frequency Offset

C. Discussion of the Results

After running the Phase Noise Effects in 256-QAM simulation in Simulink, the following results were obtained:

- Bit Error Rate (BER): 0.0001409
- Total Errors: 102
- Total Symbols Transmitted: 724,000 (7.24e+05)

These results indicate that despite the presence of AWGN and phase noise, the BER remains relatively low, suggesting that the 256-QAM system maintains good performance under the simulated conditions. The low error rate highlights the system's resilience, though further optimizations can be explored for improved reliability"

VII. CHANGE IN THE AWGN AND PHASE NOISE PARAMETERS

If you reduce the amount of AWGN while increasing the phase noise in the simulation, the impact on the Bit Error Rate (BER), total errors, and total symbols would be as follows:

A. Bit Error Rate (BER)

AWGN reduction generally improves the signal quality, reducing random amplitude variations and potentially lowering BER. Increased phase noise, however, introduces signal

distortions by affecting the phase accuracy, causing symbol misinterpretation in 256-QAM, which relies on precise phase and amplitude information. Since 256-QAM has closely spaced constellation points, phase distortions increased the BER, despite reduced AWGN.

B. Total Errors

With less AWGN, errors caused by random amplitude fluctuations decrease. But higher phase noise causes symbols to shift unpredictably in the constellation diagram, leading to more incorrect symbol decoding. As a result, total errors increased, due on the severity of phase noise.

C. Total Symbols

The total number of symbols transmitted (7.24×10^5) remains constant since it depends on the simulation runtime and system setup. However, the effective number of correctly received symbols decreases due to more errors introduced by phase noise.

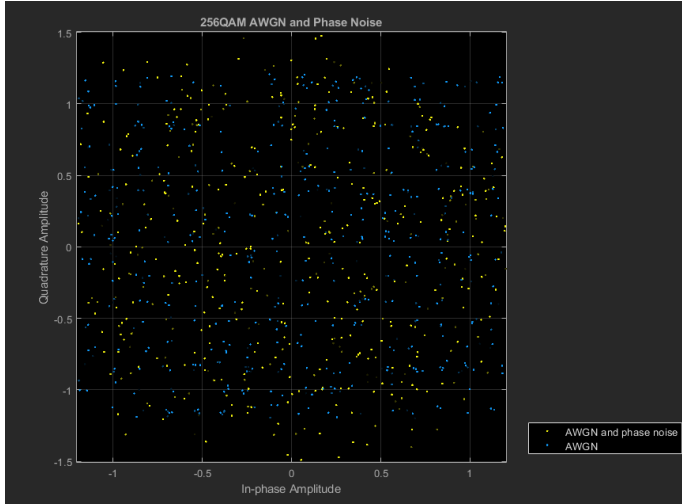


Fig. 4. 256QAM AWGN and Phase noise after reduced values of AWGN and increase in phase noise

VIII. M-ARY PHASE SHIFT KEYING (M-PSK)

M-ary Phase Shift Keying (M-PSK) is a digital modulation technique where the carrier phase is varied among M distinct values, each representing a unique symbol that encodes $\log_2(M)$ bits. This method achieves spectral efficiency by allowing multiple bits per symbol while maintaining a constant envelope, which is beneficial for power-limited systems. However, as M increases, the angular separation between symbols decreases, making the system more susceptible to phase noise and requiring a higher signal-to-noise ratio to maintain low error rates. In a laboratory setting, analyzing M-PSK helps illustrate the balance between data rate enhancement and the trade-offs in noise resilience, a key consideration in designing robust digital communication systems.

A. Replacing the QAM with M-PSK Modulation Scheme

Replacing QAM with modulation schemes like M-PSK in a Simulink simulation involves substituting the QAM modulator and demodulator blocks with their M-PSK counterparts and adjusting system parameters accordingly.

Unlike QAM, which conveys information through variations in both amplitude and phase, M-PSK encodes data solely in the phase, resulting in a constant envelope signal that is more robust in power-limited or nonlinear channel conditions. This change is evident in the constellation diagram—where M-PSK symbols are evenly spaced on a circle—and impacts system performance, as M-PSK is typically more sensitive to phase noise.

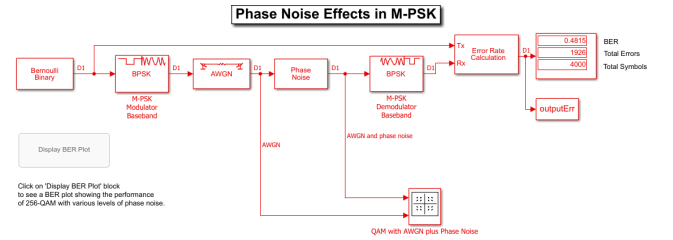


Fig. 5. Block diagram on simulink

By comparing metrics such as the bit error rate (BER) under similar noise conditions, the simulation highlights the trade-offs between spectral efficiency and noise resilience inherent in different modulation schemes.

B. Constellation Diagram of M-PSK Modulation Scheme

The M-PSK constellation diagram features M evenly spaced points around a circle, each representing a unique phase. This constant envelope layout ensures all symbols have the same amplitude, but as M increases, the reduced angular separation between symbols makes the system more vulnerable to phase noise and errors.

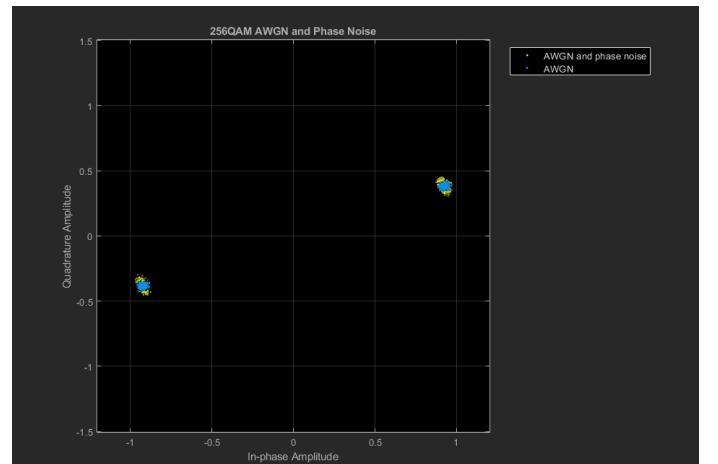


Fig. 6. Constellation Diagram of M-PSK Modulation Scheme

C. Change in the AWGN and Phase Noise Parameters

In an M-PSK modulation scheme, reducing AWGN improves the signal quality by minimizing random amplitude and phase fluctuations, which helps maintain clear, distinct phase positions in the constellation diagram. However, increasing phase noise has a detrimental effect because M-PSK encodes information solely in the phase; higher phase noise disrupts this critical phase accuracy, leading to greater symbol misinterpretation and higher bit error rates despite the lower AWGN level.

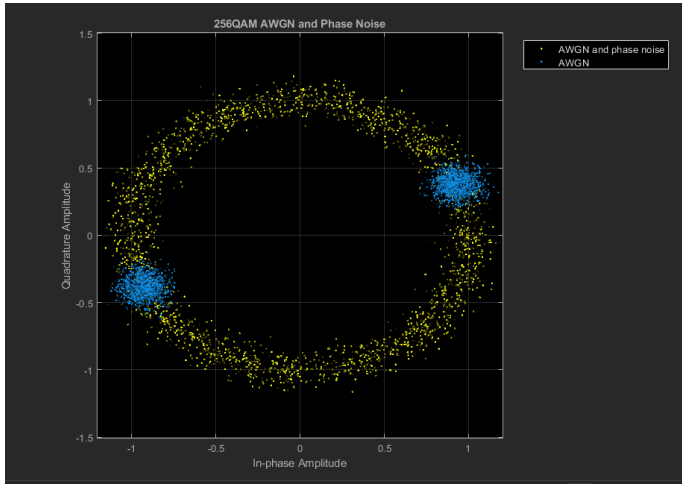


Fig. 7. Constellation Diagram of M-PSK Modulation Scheme after change in Parameters

Reducing AWGN in an M-PSK system minimizes random fluctuations, keeping the constellation points tightly clustered around their ideal phase positions. However, increasing phase noise disrupts the phase accuracy, causing the points to spread along the circle, which results in overlapping symbols and higher error rates.

CONCLUSION

In conclusion, this lab exercise provided a comprehensive exploration of digital modulation schemes using Simulink, comparing the performance of 256-QAM and M-PSK under varying channel conditions. The simulation highlighted that while 256-QAM can offer high spectral efficiency, it is sensitive to noise, as evidenced by its constellation diagram and measured BER. Transitioning to M-PSK, which encodes information solely in phase, underscored its vulnerability to phase noise—resulting in dispersed constellation points and increased error rates—despite reduced AWGN. Additionally, addressing simulation issues such as data type mismatches emphasized the need for meticulous model configuration. Overall, the study illustrated the critical trade-offs between spectral efficiency and noise resilience in digital communications, equipping us with practical insights into optimizing modulation schemes for reliable performance.

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